THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 1: **Real Numbers** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $2^2 \times 3 \times 13$	$156 = 4 \times 39 = 2^2 \times 3 \times 13$
2	**(b)** 4	$96 = 2^5 \times 3,\ 404 = 2^2 \times 101$; HCF $= 2^2 = 4$
3	**(a)** 150	LCM $= 1800 \div 12 = 150$
4	**(b)** False, since $\sqrt{2} \times \sqrt{2} = 2$ is rational	Counter-example $\sqrt{2} \times \sqrt{2} = 2$
5	**(c)** $3 + \sqrt{5}$	Rational + irrational is irrational
6	**(d)** no natural number n	$4^n = 2^{2n}$ has no factor 5, so it can never end in 0
7	**(b)** 4	LCM$(2, 4) = 4$
8	**(a)** Student A is correct	$26 = 2 \times 13,\ 91 = 7 \times 13$; LCM $= 2 \times 7 \times 13 = 182$
9	**(b)** irrational	Square root of a prime is irrational
10	**(b)** 1	Consecutive numbers are co-prime

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$3825 = 5 \times 765 = 5 \times 5 \times 153 = 5^2 \times 9 \times 17 = 3^2 \times 5^2 \times 17$	2
12	$12 = 2^2 \times 3,\ 15 = 3 \times 5,\ 21 = 3 \times 7$. HCF $= 3$; LCM $= 2^2 \times 3 \times 5 \times 7 = 420$	1+1
13	All three are primes, so HCF $= 1$ and LCM $= 17 \times 23 \times 29 = 11339$. For distinct primes, LCM = their product.	1+1
14	$336 = 2^4 \times 3 \times 7,\ 54 = 2 \times 3^3$. HCF $= 6$, LCM $= 2^4 \times 3^3 \times 7 = 3024$. $6 \times 3024 = 18144 = 336 \times 54$ ✓	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Assume $\sqrt{3} = \frac{a}{b}$ (a, b co-prime, $b \neq 0$). Then $3b^2 = a^2 \Rightarrow 3 \mid a^2 \Rightarrow 3 \mid a$; put $a = 3c$: $b^2 = 3c^2 \Rightarrow 3 \mid b$. So 3 divides both a and b — contradiction. Hence $\sqrt{3}$ is irrational.	1+1+1
16	$18 = 2 \times 3^2,\ 12 = 2^2 \times 3$; LCM $= 2^2 \times 3^2 = 36$. They meet again after **36 minutes**.	1+1+1
17	If $3 + 2\sqrt{5} = r$ (rational), then $\sqrt{5} = \frac{r-3}{2}$ is rational — contradiction. Hence $3 + 2\sqrt{5}$ is irrational.	1+1+1
	OR: $11 \times 13 \times 17 + 17 = 17(143 + 1) = 17 \times 144$; $3 \times 5 \times 7 \times 11 + 11 = 11(105 + 1) = 11 \times 106$. Each has factors other than 1 and itself ⇒ composite.	1½+1½
18	$616 = 2^3 \times 7 \times 11,\ 32 = 2^5$; HCF $= 2^3 = 8$. Maximum **8 columns**.	1+1+1
	OR: $9 = 3^2,\ 12 = 2^2 \times 3,\ 15 = 3 \times 5$; LCM $= 2^2 \times 3^2 \times 5 = 180$ min $= 3$ h. Next together at **11:00 a.m.**	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $510 = 2 \times 3 \times 5 \times 17,\ 92 = 2^2 \times 23$. HCF $= 2$, LCM $= 2^2 \times 3 \times 5 \times 17 \times 23 = 23460$. $2 \times 23460 = 46920 = 510 \times 92$ ✓ (ii) HCF always divides LCM; $380 \div 18$ is not an integer, so the claim is **false**.	3+2
	OR: Assume $\sqrt{2} = \frac{a}{b}$ (co-prime) ⇒ $2b^2 = a^2$ ⇒ $2 \mid a$, $a = 2c$ ⇒ $b^2 = 2c^2$ ⇒ $2 \mid b$: contradiction (3 marks). If $5 - 3\sqrt{2} = r$ then $\sqrt{2} = \frac{5-r}{3}$ rational — contradiction (2 marks).	3+2
20	(i) $420 = 2^2 \times 3 \times 5 \times 7,\ 130 = 2 \times 5 \times 13$; HCF $= 10$ barfis per stack. (ii) Kaju: $420 \div 10 = 42$ stacks; badam: $130 \div 10 = 13$ stacks (total 55). (iii) $260 = 2^2 \times 5 \times 13$; HCF$(420, 130, 260) = 10$ — no change.	2+2+1
	OR: (i) $850 = 2 \times 5^2 \times 17,\ 680 = 2^3 \times 5 \times 17$. (ii) HCF $= 2 \times 5 \times 17 = 170$ L. (iii) 5 times and 4 times. LCM $= 2^3 \times 5^2 \times 17 = 3400$; $170 \times 3400 = 578000 = 850 \times 680$ ✓	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 1: **Real Numbers** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $5 \times 7 \times 11 \times 13$	$5005 = 5 \times 1001 = 5 \times 7 \times 11 \times 13$
2	**(c)** 420	$2^2 \times 3 \times 5 \times 7 = 420$
3	**(a)** 36	Smallest powers: $2^2 \times 3^2 = 36$
4	**(b)** xy^2	Smallest powers: x^1y^2
5	**(b)** False; e.g. $2 + \sqrt{3}$ is irrational	Rational + irrational is always irrational
6	**(c)** $(2+\sqrt{3})(2-\sqrt{3})$	$(2+\sqrt{3})(2-\sqrt{3}) = 4 - 3 = 1$
7	**(a)** Student A is correct	Co-prime ⇒ HCF $= 1$, LCM $= 8 \times 9 = 72$; B found the HCF
8	**(c)** 4	$144 = 2^4 \times 3^2$
9	**(a)** 338	$26 \times 169 \div 13 = 338$
10	**(b)** always irrational	If $r \cdot x = s$ (rational) then $x = s/r$ would be rational

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$6 = 2 \times 3,\ 72 = 2^3 \times 3^2,\ 120 = 2^3 \times 3 \times 5$. HCF $= 6$; LCM $= 2^3 \times 3^2 \times 5 = 360$	1+1
12	$= 5(7 \times 6 \times 4 \times 3 \times 2 \times 1 + 1) = 5 \times 1009$; it has a factor 5 other than 1 and itself ⇒ composite.	2
13	$6^n = 2^n \times 3^n$. A number ending in 0 must have both 2 and 5 as prime factors; 5 is absent (uniqueness of prime factorisation). So 6^n never ends in 0.	1+1
14	LCM $= 3072 \div 16 = 192$. No — HCF must divide LCM, but $380 \div 16$ is not an integer.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Assume $\sqrt{5} = \frac{a}{b}$ (co-prime). $5b^2 = a^2 \Rightarrow 5 \mid a$; $a = 5c \Rightarrow b^2 = 5c^2 \Rightarrow 5 \mid b$. Contradiction, so $\sqrt{5}$ is irrational.	1+1+1
16	Required number = HCF$(240, 1024)$. $240 = 2^4 \times 3 \times 5,\ 1024 = 2^{10}$ ⇒ HCF $= 2^4 = 16$.	1+1+1
17	If $6 + \sqrt{2} = r$ (rational), then $\sqrt{2} = r - 6$ is rational — contradiction, since $\sqrt{2}$ is irrational.	1+1+1
	OR: If $3\sqrt{2} = r$ then $\sqrt{2} = \frac{r}{3}$ is rational — contradiction. Hence $3\sqrt{2}$ is irrational.	1+1+1
18	$16 = 2^4,\ 20 = 2^2 \times 5$; LCM $= 2^4 \times 5 = 80$ min. They meet at **7:20 a.m.**	1+1+1
	OR: $28 - 8 = 32 - 12 = 20$. LCM$(28, 32) = 2^5 \times 7 = 224$. Required number $= 224 - 20 = 204$. (Check: $204 = 7 \times 28 + 8 = 6 \times 32 + 12$)	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $144 = 2^4 \times 3^2,\ 180 = 2^2 \times 3^2 \times 5,\ 192 = 2^6 \times 3$. HCF $= 2^2 \times 3 = 12$; LCM $= 2^6 \times 3^2 \times 5 = 2880$. (ii) $12 \times 2880 = 34560$ while $144 \times 180 \times 192 = 4976640$; not equal (the rule holds only for two numbers).	3+2
	OR: Assume $\sqrt{7} = \frac{a}{b}$ (co-prime) ⇒ $7b^2 = a^2$ ⇒ $7 \mid a$, $a = 7c$ ⇒ $b^2 = 7c^2$ ⇒ $7 \mid b$: contradiction. If $2 + 3\sqrt{7} = r$, then $\sqrt{7} = \frac{r-2}{3}$ is rational — contradiction.	3+2
20	(i) $896 = 2^7 \times 7,\ 672 = 2^5 \times 3 \times 7$; HCF $= 2^5 \times 7 = 224$ cm. (ii) $\frac{896}{224} \times \frac{672}{224} = 4 \times 3 = 12$ tiles. (iii) $28 \times 21 = 588$ tiles.	2+2+1
	OR: (i) $48 = 2^4 \times 3,\ 72 = 2^3 \times 3^2,\ 108 = 2^2 \times 3^3$. (ii) LCM $= 2^4 \times 3^3 = 432$ s $= 7$ min 12 s ⇒ 7:07:12 a.m. (iii) $3600 \div 432 = 8.33$ ⇒ **8 times**.	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.